

Leif Friedrich

S&C Portfolio

## Testing Protocol for NAIA Collegiate Baseball Pitchers

Tests in order of completion:

1. General non fatiguing tests
2. Max power tests

### 1) Pre-warmup to get baseline mobility scores

- a. Height, weight
  - i. Equipment: weight scale, tape measure
- b. Shoulder IR and ER degrees range of motion (looking for more than 45 degrees internal, and more than 135 degrees external)
  - i. Equipment: none
  - ii. Protocol:
    1. Set up in a half kneeling position parallel to a smooth wall. The leg facing the wall should have the knee down, and the other up. From here, with a bent elbow the athlete laterally raises their shoulder, and puts the end of their elbow against the wall. The body can now be adjusted to the torso and lower body are straight up and down.
    2. Set up the arm so that the forearm is parallel with the ground, palm facing down. This can be considered 0 degrees.
    3. For IR. Once ready, have the athlete with stiff posture, start to rotate/arc their hand toward the ground, make sure they are not compensating with any of the following: trunk flexion, trunk rotation, or shoulder girdle upward rotation. The movement should be coming solely from the shoulder joint.
    4. For ER. Once ready, have the athlete with stiff posture, start to rotate/arc their hand toward their head, make sure they are not compensating with any of the following: trunk extension, trunk rotation, or excessive shoulder girdle downward rotation (a slight amount is normal). The movement should be coming solely from the shoulder joint.
    5. When the athlete hits their maximum range without any compensation, using just your eyes measure based off of zero

how far they have rotated. For IR we are looking for ~45 degrees below 0, and for ER we are looking for 135 degrees beyond zero, or 45 degrees past 90. Record this number.

2) Max power tests – on force plates

a. Counter movement jump (CMJ)

i. Equipment: force plates

ii. Protocol:

1. Set up the plates with the center of each one shoulder width apart. Have the athlete set up with each foot in the middle of the plates, with hands not their hips.
2. When ready, have the athlete go down, jump up as high as they can, and stick the landing. Make sure to cue them to push into the ground as hard as possible, and to drive their head through the roof.
3. Give 15-30 seconds in between each trial, and repeat for 3 total jumps.
4. Multiple measurements/scores can be taken from this test. Primarily we want to know jump height (cm), RSI-mod, and left-right asymmetries. Record the average over the 3 trials.

b. Isometric mid thigh pull

i. Equipment: force plates, chain, straight bar

ii. Protocol:

1. Set up the plates with the center of each one shoulder width apart. Have the athlete set up with each foot in the middle of the plates. Measure the bar so that the chain is tight when the bar is around the mid part of the athlete's thigh.
2. Cue the athlete to have a straight back, apply tension to the bar, and when they are ready, to rip the bar out of the ground as fast as possible. Repeat 1-2 times total.
3. With this test we are looking for peak force (N). Record the average over all trials.

c. Grip strength test

i. Equipment: hand grip dynamometer

ii. Protocol:

1. Have the athlete grip the dynamometer firmly, and record the max force applied (N).